

Skip-YOLO: Domestic Garbage Detection Using Deep Learning in Complex Multi-Scenes

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Abstract

Waste management, which is a global challenge due to rapid urbanization, population growth, and limited waste disposal or storage opportunities, can be improved through the classification of domestic waste, thereby reducing the complexity of recycling and material recovery and improving environmental sustainability. Standard manual sorting techniques can be inefficient with large amounts of input, since they depend on human labor and human judgment.

Most of the contemporary deep learning-based object detectors do not perform well on garbage detection in real-world settings, as there are variations in the background, occlusions, too many objects in close proximity to one another, and the high correlation between different types of trash in a scene, such as on the road or in public trash bins.

To overcome these limitations, Skip-YOLO, a variant of the YOLOv3 model, is proposed. Skip-YOLO uses dense convolutional blocks, large convolutional kernels, and multi-scale feature fusion to extract features. With these changes, Skip-YOLO is able to leverage broader contextual information, and experimental results show that Skip-YOLO outperformed customary YOLO models in terms of accuracy and generalization ability in highly degenerate scenarios, which opens up the possibility of using YOLO models in practical scenarios such as waste treatment systems.

Keywords

Deep Learning, Object Detection, YOLOv3, Garbage Detection, Computer Vision

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1 Introduction

As more waste is generated globally, waste management is an essential aspect of sustainable development. Improper waste disposal leads to pollution, adverse health effects, and inefficient recycling. Computer vision-based automated waste detection systems for smart cities can reduce human intervention and improve efficiency in the recycling process, while also reducing the adverse effects of improper waste disposal.

Customary methods to sort garbage by hand are inefficient, unstable and not suited for mixed waste with large volumes. The state of the art deep learning object detection algorithm YOLOv3 has not worked well for garbage sorting tasks because of the complexity and diversity of real world data sets.

There also may be meaningful confusion between different wastes due to the cross-similarity in color, surface texture or morphology - for example, a plastic bottle may simulate a glass bottle reasonably well. Other factors that lead to poor object detection performance include cluttered backgrounds, occlusions, small and/or scattered objects. Furthermore, deep neural networks are often trained on limited data, which leads to overfitting and low generalization performance.

Waste categorization systems, a growing aspect of smart city infrastructure, have the potential to eliminate the need for human involvement through automation. Governments and environmental agencies are increasingly focusing on Artificial Intelligence based waste monitoring systems for recycling accuracy and reducing pressure on landfills. Thus research in this field is of technical and social importance for the environment as well as providing technological innovation.

2 Proposed Method: Skip-YOLO

Skip-YOLO was developed as an extension of YOLOv3, enabling it to learn features more effectively and improve performance in the challenging task of garbage detection with an important amount of real-world data variability.

2.1 Key Improvements

1. Large Convolution Kernel (7×7) Using a large convolution kernel increases the receptive field of the network and thus provides more context. This is especially useful when the images contain a lot of background clutter, with objects that are occluded or masked with background noise. The model learns more contextual information, leading to improved performance on similar tasks.

2. Dense Convolution Blocks Here, each layer is densely connected to all preceding layers. This mechanism encourages feature re-use, improves gradient propagation, and reduces information loss to every layer, thus augmenting the amount of information available for learning and resulting in more stable and less-overfitting models.

3. Multi-Scale Feature Fusion As multi-scale feature fusion can obtain the feature maps from different levels, it can detect both small and large-size waste objects in the same scene, which is especially important for arbitrary waste detection due to the variety of shape and size.

4. CBL Module (Convolution, BatchNorm, and Activation) The CBL module improves training stability, convergence rate, reduces internal covariate shift and compresses feature maps thereby providing increased model efficiency.

3 Methodology

The pipeline consists of three stages.

3.1 Data Preparation

Images were taken from both indoors and outdoors, including streets, parks, dormitories, and classrooms in order to have a dataset that captures various environments within which the object detection model would be used. All images were resized to a uniform 416x416 pixels.

The dataset was divided into 80% training, 10% validation and 10% testing. This data split allows for proper evaluation of the model and prevents overfitting. To further increase the classification difficulty, other classes of waste were added such as recyclable, non-recyclable and hazardous waste.

3.2 Feature Extraction

The deeper levels in these blocks retain spatial and contextual information allowing us to combine this information with higher-level knowledge from the more shallow layers, gradually allowing us to extract finer and more high-level semantic features from the images.

3.3 Detection

Detection layers of YOLO predict bounding boxes and waste classifications. YOLO performs classification and bounding box regression simultaneously, therefore it is highly suitable for real-time applications.

In post-processing, steps like non-maximum suppression (NMS) further improve predictions by discarding overlapping bounding boxes to retain only the most likely, thereby reducing false positives in detected regions.

In addition, a process called Non-Maximum Suppression (NMS) may be applied to remove duplicate bounding boxes and improve the accuracy of the final prediction. NMS eliminates a number of overlapping bounding boxes from the detected boxes to reduce the number of predictions for a single object. The entire pipeline ensures that both detection accuracy and localization accuracy are maintained through the process.

4 Experimental Results and Key Findings

The model also was evaluated on a dataset of real-world domestic waste to understand its performance.

Key Results:

- mAP50 improved by 22.5% over YOLOv3
- Recall increased by 18.6%
- Achieved 90.38% accuracy
- Exceeds YOLOv5, YOLOv7, and DETR in complex scenes

Observations:

- Good performance in dense and multi-object environments
- Improves robustness under different lighting conditions
- Less misclassification between similar waste types
- Better generalization on unseen data

The model is also more resistant to harsher real-world conditions such as rain, shadow and uneven lighting of the test images. This result indicates that Skip-YOLO generalizes well in real-world images, and not only in synthetic datasets. Another important result to consider is the reduction of false positives for visually similar waste types, that is a common challenge in YOLO-based architectures.

5 Critical Analysis

Skip-YOLO shows that simply improving the convolutional neural network architecture used by YOLO can improve object detection performance without modifying YOLO itself.

The model generalizes well to other real-world waste management systems but may be unsuited for mobile and edge-computing applications due to its relatively high computational cost.

Despite Skip-YOLO showing the advantages of combining dense connections with multi-scale feature fusion for higher accuracy and robustness, scalability remains an issue, as deeper models can demand higher computational power and memory bandwidth.

Furthermore, such techniques typically require large datasets which renders their training expensive and time consuming and the data annotation laborious. Despite these disadvantages, the model has been used for industrial applications where precision is prioritized over speed.

6 Conclusion

Skip-YOLO is another extension to garbage detection in complex environments, which improves feature representations by using a combination of dense blocks and multi-scale fusion, outperforming YOLOv3.

The results were satisfactory and the model showed promise in practical smart waste management applications. Despite a relatively high computational cost, the additional accuracy and robustness of the model suggest it would be practical to implement.

Future work seeks to reduce model size and computational cost for real-time performance. Investigating the use of lightweight models and unsupervised learning could also improve performance.

References

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